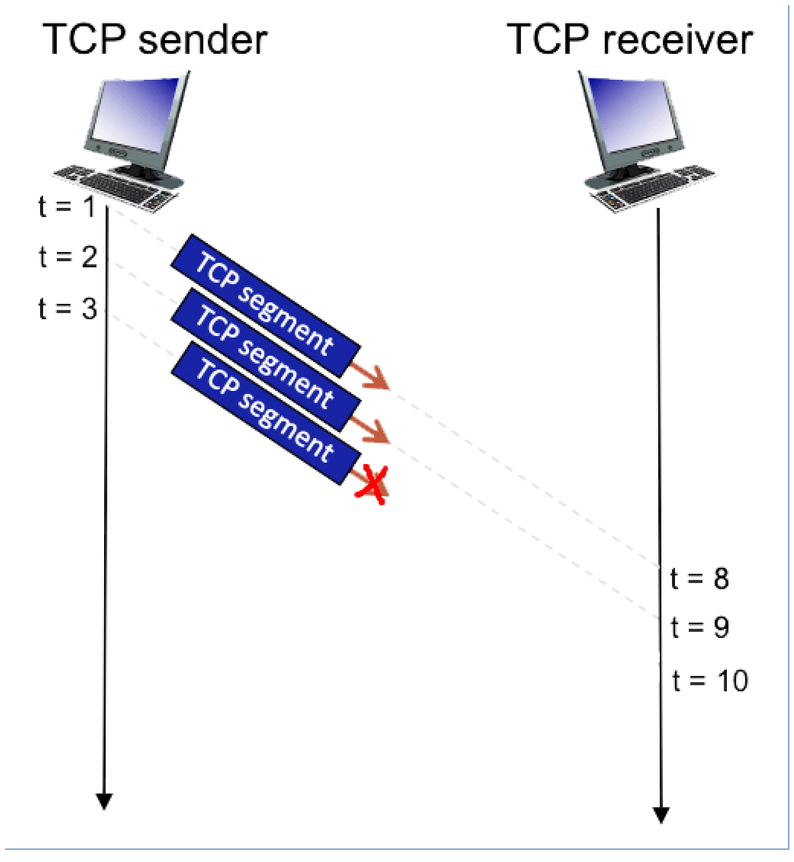
INTERACTIVE END-OF-CHAPTER EXERCISES

TCP SEQUENCE AND ACK NUMBERS WITH SEGMENT LOSS

Consider the figure below in which a TCP sender and receiver communicate over a connection in which the sender->receiver segments may be lost. The TCP sender sends an initial window of 3 segments. Suppose the initial value of the sender->receiver sequence number is 252 and the first 3 segments *each* contain 856 bytes. The delay between the sender and receiver is 7 time units, and so the first segment arrives at the receiver at t=8. As shown in the figure below, 1 of the 3 segment(s) are lost between the segment and receiver.



QUESTION LIST

1. Give the sequence numbers associated with each of the 3 segments sent by the sender. Format your answer as: a,b,c,...

2. Give the ACK numbers the receiver sends in response to each of the segments. If a segment never arrives use 'x' to denote it, and format your answer as: a,b,c,...

SOLUTION

1. The sender's sequence numbers are: 252,1108,1964

2. The receiver's ACKs are: 1108,1964,x

QUESTION 1 OF 2

Give the sequence numbers associated with each of the 3 segments sent by the sender. Format your answer as: a,b,c,...

Answer

CHECK | HINT | SKIP

We greatly appreciate the work of John Broderick (UMass '21) in helping to develop these interactive problems.

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